

RAJDEEP GILL

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EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

May 2026

Winnipeg, MB

- GPA: 4.45/4.5
- Relevant Coursework: Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming

EXPERIENCE

Undergraduate Research Student

May 2024 - Dec 2024

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Designed and trained a convolutional neural network (CNN) to predict biological cell viability from 10,000+ diffraction pattern images, achieving 85% accuracy and showcasing the potential for non-invasive cell viability assessment.
- Enhanced particle diffraction centroid detection using edge detection and clustering algorithms, increasing success rate from 75% to 92%, providing more accurate data for the CNN model.
- Created a Python-based data visualization tool with an interactive GUI to process, filter, and display diffraction patterns.
- Integrated neural network predictions with Shapley value visualizations into the data visualization tool, enabling researchers to interpret model decisions and identify key features influencing predictions.
- Implemented a scalable SQL database to store and manage 100,000+ optical diffraction records, allowing for efficient data retrieval and a version-history system for data analysis.

Undergraduate Research Student

May 2023 - Aug 2023

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Applied dimensionality reduction techniques (t-SNE, PCA) to compare and cluster datasets, uncovering key correlations and guiding future research efforts
- Accelerated data analysis by 90% through parallel processing, significantly improving workflow efficiency
- Created an oscilloscope interfacing program for a separate lab project, providing easier data collection
- Delivered weekly updates in team meetings, sharing progress and gathering feedback from professors and students

PROJECTS

Google Docs Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing
- Utilized Convex to store and sync document data, allowing multiple users to edit the same document simultaneously
- Implemented organization-wide document sharing and user permissions to control access to documents

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js

- Created an expense and income tracking dashboard with user authentication and real-time data visualization
- Implemented a RESTful API to handle user data and transactions, and a PostgreSQL database to store user information

Discord Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Discord clone with user authentication, real-time messaging, video and voice chat functionality
- Utilized Next.js for server-side rendering, socket.io for real-time messaging with a polling fallback

Microprocessor Communication System | C, Java

- Designed and implemented a server using the TCP/IP stack in C, handling multiple concurrent connections
- Developed a Java-based proxy server to interface between an Android application and the microprocessor server
- Designed an Android application to allow users to control the LEDs, monitor temperature and button states

AWARDS

Engineering Academic Excellence Award - Computer Engineering

2024

NSERC Undergraduate Student Research Award

2023, 2024

President's Scholarship

2021-2024

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, R, SQL, MATLAB

Frameworks & Libraries: React, Next.js, TensorFlow, PyTorch, OpenCV,

Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ